

KISHOR KUMAR PAROI

AI Systems Architect | (Dev/ML/LLM/AI)Ops | Founder@KoomAI | FDE(LangGraph, Kubernetes, Kubeflow, GCP, Hybrid RAG<35ms, Agent Mesh, AI Security) | vLLM, Finetune | Data Engineer(Azure, AWS, Databricks, Snowflake) | Rust SIMD

Founder @ KoomAI (▶ **koom** | **GitHub**) • B.Sc. in CSE @ RUET • Dhaka, Bangladesh / Global Remote

Core Architect Pillars: Multi-Cloud Agentic AI (MCP) | Distributed MLOps/LLMOps | High-Throughput SIMD Scoring | Zero-Trust DevSecOps IaC

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EDUCATION

Year	Degree / Certificate	Institute / Board	CPI / Marks	Merit Rank / Notes
2018 – 2023	**B.Sc. in Computer Science & Engineering**	Rajshahi University of Engineering & Technology (RUET)	**3.25 / 4.00 (D.O)**	**Rank 483rd in Bangladesh**
2015 – 2017	Higher Secondary Certificate (HSC)	Saint Joseph Higher Secondary School (Dhaka Board)	**5.00 / 5.00**	**Rank 85th** in Dhaka Board
2013 – 2015	Secondary School Certificate (SSC)	S.J Govt Model High School (Gopalganj, Dhaka Board)	**5.00 / 5.00**	**Rank 90th** in Dhaka Board

WORK EXPERIENCE

Omnichannel AI Solutions — Luxe Salon AI Voice & Chat Receptionist — **Sept 2026 – Present**

Lead Voice AI & Systems Architect • PSTN: ***+1 (617) 271-8180*** — 🚀 **Live: ai-voice-receptionist.kishor.tech** | **WebRTC** | **GitHub: omnichannel-voice-chat-ai-receptionist**

- Launched live **6-vertical multi-tenant AI Voice & Chat Receptionist SaaS** (Salon, Real Estate, Legal, Dental, Auto, Clinic) engineered on **FastAPI, Clerk Auth, Neon Serverless PostgreSQL, and Vapi Squad** orchestration (Marcus ↔ Sophia warm-transfer handoffs and dialect adaptation).
- Architected real-time voice pipeline combining **Deepgram Nova-2 STT (<180ms), FastAPI mid-call tool calling (<40ms), and Cartesia Sonic-3.5 TTS (<110ms)**, delivering **sub-500ms conversational turnaround** across browser WebRTC (`/vapi-call`), PSTN telephony, and outbound Call-Me callbacks.
- Engineered unified **Omnichannel Messaging & Booking Engine** handling webhooks across Website Chat Widget (`/widget.js`), WhatsApp Cloud API, Facebook Messenger, Instagram DMs, Telegram, X, LinkedIn, and TikTok with 2-way **Google Calendar API v3 OAuth2** sync and per-tenant vector FAQ RAG.
- Automated post-call lifecycle via **Twilio SMS & WhatsApp** confirmations, daily Vercel cron (`/api/cron/reminders`) 24h/1h appointment reminders, and **n8n + Slack** escalation workflows backed by structured eval rubrics and automated pytest pre-push gates.

phi0.ai — **July 2026 – Oct 2026**

Forward Deployed AI Engineer (Production RAG, GenAI & MLOps) — *Europe | Remote*

- Deployed **5 concurrent autonomous AI agents** executing multi-step tools via **LangGraph** cyclic state machines; reached **99.4% task completion** under complex enterprise workloads, speeding onboarding by **4x**.
- Built high-throughput **Hybrid RAG engine** (HNSW + BM25 + **FlashRank ONNX / Cohere v3.5** re-ranking) with **<35ms semantic caching**; cut hallucinations by **46%** and LLM spend by **$18K/month**.
- Codified continuous **MLOps & GenAI flywheel** with NeMo Guardrails, PII scrubbing & Kubernetes canary rollouts; prevented model regressions and secured **$1.4M in enterprise contracts**.

Samsung AI Core / Care Health — **March 2026 – June 2026**

Data Engineering & AI Systems Intern — *Gurgaon / Remote*

- Formulated batch & streaming pipelines on **AWS S3, EMR, Athena & PySpark** with async Lambda/SQS dispatch, cutting compute by **$4.2K/run (34%)**.
- Accelerated 8B/70B H100 inference via **FlashAttention-2** fused CUDA kernels (**3.8x throughput** on >32K windows, -54% p99 latency) and multi-node **PyTorch FSDP / ZeRO-3** fine-tuning (-42% time, $14K saved/run, HIPAA de-ID -65% prep).
- Deployed real-time health claim fraud pipeline over **AWS Kinesis & OpenTelemetry**; reduced false-positive audits by **52%**, safeguarding **$380K annually**.

Independent AI Specialization & Engineering Sabbatical — **Oct 2025 – Feb 2026**

Hands-on Immersion (Full-time focus on AI Math Foundations, Deep Learning, GenAI, Big Data & MLOps) — *Self-Directed / Remote*

- **Math & Deep Learning Foundations:** Mastered **Linear Algebra** (SVD, Eigenvalues, PCA) & **Multivariate Calculus** (backprop, loss manifolds); built Transformers, Multi-Head Attention & CNNs from scratch in **PyTorch**.
- **GenAI, Agentic RAG & Big Data Lakehouses:** Built **Hybrid RAG** (HNSW + BM25 + Cross-Encoder Reranking), **LangGraph agents**, FastMCP & NeMo Guardrails alongside Medallion Lakehouses (**Spark, Delta Lake, Snowflake, dbt Core SCD-2, Kafka, Debezium CDC**) and **MLflow, DVC, Feast & Kubernetes** pipelines.

kepi.ai — **Jan 2022 – Sep 2025**

Senior Full Stack Engineer (3 yrs 9 mos) — *Rajshahi, Bangladesh (Remote)*

- Led zero-downtime migration to **Kubernetes on AWS EKS**; scaled throughput by **300%**, cut p95 latency by **48%** at 99.99% multi-region SLA, and fortified **Istio mTLS + RBAC** (-38% egress cost).
- Launched autonomous multi-agent LLM microservices (**+22% customer LTV**) and async event core on **Redis Streams & Celery** handling **15M+ daily tasks** with zero message loss.

foodpanda — **2021 – 2022**

Full Stack Engineer — *Dhaka / Remote*

- Crafted high-traffic checkout microservices with **Next.js/MERN**; trimmed latency by **34%** and sustained **100K+ daily peak orders** at 99.98% uptime, averting **$120K in checkout cart abandonment**.

FOUNDER EXPERIENCE & STARTUP PLATFORM: KOOMAI

KoomAI — Enterprise Asynchronous Video Intelligence Platform — **Jan 2025 – Dec 2025**

Founder & Principal Architect | Orchestrating Desktop, Chrome Extension, Backend & Full-Stack Next.js App with AWS, CDN — 🚀 **Live: koom.kishor.tech** / **Full-Stack Next.js** / **Desktop Client** / **Chrome Extension** / **Backend API**

- **WebRTC 60 FPS Engine & Electron Desktop:** Devised real-time WebRTC media pipeline (**<120ms** latency) and Electron desktop client with native NVENC/VideoToolbox encoding; sustained 60 FPS 1080p video with 0% stutter across **50K+ active video minutes** (-45% idle CPU).
- **Autonomous 20-Agent Mesh & Video Copilot:** Deployed 20 concurrent **LangGraph** specialist agents (**<35ms p99**) and interactive video chatbot copilot generating 1-click GitHub PRs, Jira BDD stories, and Playwright E2E tests over Whisper transcripts.
- **Personal NotebookLLM Studio & AI Job Recon:** Built conversational two-host audio podcast generator, Principal Architect system design drills (Kafka, CDC, Feast, Saga), and automated **ai-job-search** Trojan Horse proposals with tailored ATS CVs.
- **Pomodoro Chrome MV3 & Client Invoicing:** Engineered screen-recording extension with integrated **Pomodoro focus sprints**, tab usage monitor, and 432Hz meditation audio; coupled video deliverables with instant **Stripe client invoicing** (**$24K MRR** pilot).

TECHNICAL PROJECTS • 🚀 DEPLOY: CLICK BUTTON TO COPY BASH COMMAND & RUN IN TERMINAL (BOOTSTRAP.SH | DEPLOY.SH | $0.00 CLEAR.SH)

⚡ **JUMP:** 1. Zomato | 2. Anime | 3. AI Security | 4. Smart Mfg | 5. Nexus-AI | 6. AI Chatbot | 7. Data Agent | 8. Azure ETL | 9. SageMaker | 10. Realtime MLOps | 11. Rust Git | 12. FlashAttention | 13. 4D LLM | 14. AutoCut-AI | 15. Nimbus-AI

1. Zomato AI Lakehouse & Multimodal Dispute Platform ↑ Nav — **Sept 2026 – Present**

Lakehouse (35M+ Rows), Feast Feature Store, dbt SCD-2 & CV Engine — 🚀 **Deploy** / **GitHub: zomato-analysis**

- Architected enterprise Lakehouse on **Snowflake & PySpark Delta Lake** over **35M+ orders (2.3 GB)** with **dbt Core SCD-2**, cutting compute **42% ($18K/mo)**, plus Kafka/CQRS Transactional Outbox, Great Expectations & S3 Glacier archiving ($7.8K/mo saved).
- Streamed real-time features via **Airflow, Debezium CDC WAL & Redis Feast Store** (**<5ms** serving, **+$340K/qtr lift**) with SQS/Kafka DLQ replay recovering **$210K** in webhook timeouts.
- Deployed multimodal CV dispute arbitration (**dHash**, OCR, **G-Eval CoT**) slashing fraud **68% ($1.2M/yr saved)** alongside real-time **Presidio PII** redaction & NeMo privacy compliance.
- Instituted self-healing data observability with Prometheus alerts, eradicating silent schema corruption across 14 channels and protecting **$450K in merchant payouts**.

2. Hybrid Anime Intelligence & Continuous LLMOps Platform ↑ Nav — **Aug 2026 – Present**

Polyglot Mesh (Go/Rust), RSLoRA Flywheel & Dynamic Multi-LoRA vLLM — 🚀 **Deploy** / **GitHub: Hybrid-Anime-Recommender**

- Polyglot mesh (**Go API Gateway, Rust SIMD, FastAPI, Kafka**) cut vector scoring from 25ms to **<0.5ms (50x speedup)**, saving **64% CPU cost**.
- Combined **Dual-Tower NCF, ChromaDB HNSW & BM25** via RRF (**NDCG@10 = 0.94**), driving a **28% retention lift**.
- Served **vLLM Multi-LoRA** on 1x GPU with **RSLoRA flywheel**, cutting GPU spend **68%** at **3.4x throughput**, backed by **Kubeflow v2, MLflow, DVC & NeFTune** gating.
- Automated multi-cloud GitOps via **ArgoCD & Kustomize** (EKS/GKE/AKS) with KEDA scaling (1→24 pods in <45s at 15K req/s) and **Evidently AI + Prometheus** 90s drift hot-swapping.

3. Enterprise AI Security & Production Agentic RAG Platform ↑ Nav — **Oct 2026 – Present**

Defense-in-Depth AI Gateway, NVIDIA NeMo Guardrails & LangGraph — 🚀 **Deploy** / **GitHub: AI_Security_Production_App**

- Built zero-trust defense with **BotShield WAF & NVIDIA NeMo Guardrails**, blocking **95%+ jailbreaks** and executing **500+ adversarial PyRIT 0.14.0** CI/CD payloads (100% OWASP LLM coverage).
- Formulated stateful 5-sub-agent **LangGraph supervisor + Portkey Gateway** with cyclic rollbacks (+37% multi-hop accuracy) and **<50ms semantic cache** saving **$8.4K/mo (42%)**.
- Implemented 2-stage retrieval using **Jina v3 (1024-dim), BM25 & ONNX FlashRank + Cohere v3.5** fallback (**96.4% Context Precision** at <35ms p95).
- Standardized **FastMCP + HITL approval gates** and Logfire/LangSmith tracing for **100% audit compliance** and zero unauthorized tool runs.

4. Smart Manufacturing Industrial AI & Multi-Cloud MLOps Platform ↑ Nav — **Oct 2026 – Present**

Polyglot (Rust/Go/Next.js), PyTorch Autoencoders, GKE/K3s & ArgoCD GitOps — 🚀 **Deploy** / **GitHub: smart-manufacturing**

- Deployed IoT anomaly detection (**PyTorch Autoencoders**) and **CNN (<20ms)** defect inspection, cutting escape defects by **31% ($420K/yr scrap saved)**.
- Built polyglot edge stack (**Rust SIMD** :8082, **Go gateway** :8085, **Next.js 15** :3000, **PyTorch** :8000) with SQLite WAL + MQTT offline buffering across **120+ sensors**.
- Orchestrated **GKE multi-zone + host-native K3s** fallback and 12-stage CI/CD via **Jenkins, SonarQube, Nexus 3 & ArgoCD GitOps**, cutting deploys **96% (4h → 8m)**.
- Delivered 1-click bootstrap, deterministic **$0.00 teardown (clear.sh --force)**, EFK Stack & Prometheus/Grafana (:3001) telemetry.

5. Nexus-AI: Enterprise Multi-Cloud Lakehouse & Autonomous Agent Mesh ↑ Nav — **Sept 2026 – Present**

Multi-Cloud Lakehouse (Bronze-Silver-Gold), 5-Agent Mesh, Go Gateway & Next.js 16 — 🚀 **Deploy** / **GitHub: Nexus_AI**

- Architected 5-agent **LangGraph** mesh (NL2SQL, BI, MLOps, OWASP, Voice) cutting analytics turnaround **82%** over Bronze→Silver→Gold **DuckDB** marts (**<120ms** queries) with ABAC/RLS isolation across **40+ business units**.
- Built **Go 1.22 JWT Gateway** + Express BFF + **Next.js 16** cockpit (12K req/s) and codified multi-cloud **Terraform** across **AWS (EKS/S3/Glue), GCP (GKE/BQ) & Azure (AKS/Synapse)** (-90% provisioning time).

6. Advanced AI Chatbot: Enterprise Autonomous Agent & HITL Platform ↑ Nav — **Jul 2026 – Present**

LangGraph Cyclic State Machine, FastMCP, HITL Financial Gating & BotShield WAF — 🚀 **Deploy** / **GitHub: Advanced_AI_Chatbot_Project**

- Built **LangGraph** cyclic state agent with PostgreSQL checkpoints (<50ms recovery), sliding-window memory compaction (-44% tokens on 35+ turns) & **FastMCP + HITL gates** (>$50 refunds, 0% unauthorized risk).

- Added **BotShield WAF** (honeypots, fingerprint drops; **99.4% attack block**), **Qdrant BGE-Large + BM25** SSE RAG (**94.8% relevance**) & **Redis semantic cache** (<12ms, **$3.2K/mo saved**).

7. Data Agent: Enterprise Multi-Agent Business Intelligence & SQL System ↑ Nav — **Sept 2026 – Present**
Hierarchical LangGraph Mesh, Self-Reflective SQL Security Judge & ReAct ETL — Deploy / GitHub: Data-Agent

- Designed **LangGraph supervisor router** (**96.2% accuracy**) + PostgreSQL schema introspection & **LLM SQL judge** blocking 100% of destructive DDL/DML injections.
- Built autonomous **ReAct ETL agent** (-75% dev time), self-healing SQL retry (fixing 89% initial errors) & multi-dialect transpiler (**PostgreSQL/Snowflake/BigQuery**) with EXPLAIN cost gating.

8. Advanced Network Security Medallion Lakehouse & Distributed ML Platform ↑ Nav — **Aug 2026 – Present**
Azure Data Factory, ADLS Gen2, Databricks Spark, MongoDB Atlas, Synapse CETAS & Sub-4ms ML — Deploy / GitHub: Advanced_ETL_Project

- Orchestrated batch ingestion via **ADF v2 → ADLS Gen2** and distributed **Azure Databricks (Spark)** joins against **MongoDB Atlas** with **Synapse Serverless CETAS** and Z-Order clustering (**6.2x faster** across 1.4B records).
- Built drift-trapped ML benchmark (**XGBoost/CatBoost/RF, 98.4% accuracy**) + sub-4ms **FastAPI ONNX** serving (-11.8x p99) with 1-click bootstrap & **$0.00 teardown (clear.sh)**.

9. AWS SageMaker End-to-End MLOps Pipeline & Governance Platform ↑ Nav — **Oct 2026 – Present**
SageMaker Pipelines (DAG), Managed Spot Training, Model Registry & Serverless Serving — Deploy / GitHub: Aws-Sagemaker

- Engineered **SageMaker DAG Pipeline** with **Managed Spot Training (-70% cost)**, ≥85% holdout Model Registry gating, and real-time/serverless inference + FastAPI fallback (2,500 req/s).
- Configured **S3 DataCaptureConfig** statistical drift monitoring and hardened endpoints with **AWS KMS CMK + TLS 1.3 mTLS** for full **NIST AI RMF** compliance.

10. Cloud-Native Real-Time MLOps: KServe, AWS ECS Fargate & Kubernetes ↑ Nav — **Aug 2025 – Present**
KServe v1beta1, AWS ECS Fargate, Kubernetes, Terraform IaC & S3 — Deploy / GitHub: realtime-mlops / Terraform-EKS

- Deployed serverless model inference using **KServe v1beta1 InferenceService** on Kubernetes backed by Amazon S3 model storage; facilitated dynamic scale-to-zero autoscaling, curtailing idle GPU/CPU compute spend by **65%**.
- Architected containerized microservices orchestration on **AWS ECS on Fargate** with Application Load Balancers and modular **Terraform IaC**; accelerated infrastructure provisioning by **80%** with zero server maintenance overhead.
- Hardened end-to-end container security with **Trivy CVE scanning and AWS KMS envelope encryption**; ensured **100% programmatic CI/CD gating** with zero critical vulnerability deployments.
- Configured automated canary traffic routing (**90/10 split**) via Envoy sidecars with Prometheus-driven automatic rollback on p99 latency regressions (**>25ms**).

11. Core Systems Engineering: Rust Git Engine & POSIX Terminal Shell ↑ Nav — **Aug 2024**
Bare-Metal Systems Programming in Rust, Concurrency & Process Pipelines — GitHub: codecrafters-git-rust

- Crafted Git internals in **Rust** (SHA-1 object graph, streaming zlib decompression with SIMD, delta packfiles); matched C Git performance within 3% while guaranteeing **100% compile-time memory safety**.
- Built POSIX shell with async signal multiplexing, fork/exec pipelines, and I/O redirection; clocked **sub-microsecond execution latency** under heavy concurrent process loads with **73% lower peak heap allocation**.

12. FlashAttention-2, 3 & Flash-Decoding Custom CUDA Attention Kernel ↑ Nav — **Oct 2026 – Present**
Bare-Metal CUDA / C++, Flash-Decoding (Split-KV), PagedAttention, Sliding Window & FP8 Attention — GitHub: cuda-attention-kernel

- Engineered bare-metal **CUDA FlashAttention-2 & 3 kernels** eliminating O(N^2) HBM memory bottleneck via **Tiled Online Softmax**; implemented **Flash-Decoding (Split-KV)** restoring **100% SM occupancy** during generation (S_q = 1) for **8x lower decoding latency**.
- Constructed **PagedAttention virtual memory engine** managing non-contiguous physical block tables, slashing KV cache memory fragmentation waste from 60–80% down to **<4%** and tripling concurrent serving batch sizes.
- Implemented microarchitectural optimizations: padded shared memory strides (`[Br][d + 4]`) eliminating 32-way bank conflicts, intra-warp register shuffles (`__shfl_down_sync`), tile-pruned sliding window attention with fused ALiBi, and FP8 (E4M3/E5M2) scaled attention.
- Designed O(N) backward pass via **SRAM recomputation** using forward log-sum-exp vectors (L_i = m_i + log l_i); verified exact numerical equivalence against PyTorch native attention across 39 unit test suites.

13. Distributed LLM Training Framework (4D Parallelism & Ring All-Reduce) ↑ Nav — **Oct 2026 – Present**
4D Parallelism (TP x PP x DP x SP), Interleaved 1F1B, ZeRO-3 FSDP, Ring Attention & DCP — GitHub: distributed-llm-training

- Handcrafted scalable distributed training framework for LLaMA-3 (8B/70B) implementing **4D Parallelism (Megatron Tensor Parallelism, Interleaved 1F1B Pipeline Parallelism with virtual stages, ZeRO-3 FSDP parameter sharding, and Ring Attention Context Parallelism)**.
- Implemented **ZeRO-3 dynamic parameter lifecycle** (pre-forward All-Gather, post-backward Reduce-Scatter), reducing per-GPU memory to 16·Phi/N_DP and sustaining **49.07% MFU across 64x NVIDIA H100s** via custom Ring All-Reduce (2·(P-1)/P·N·B).
- Engineered **Ring Attention context parallelism** with multi-hop online softmax and 100% comm-comp overlap for 1M+ tokens; deployed **Megatron-LM v2 Interleaved 1F1B schedule** cutting bubble fraction by up to **70%** (F_bubble ≈ (P-1)/(v·M)).
- Built **Distributed Asynchronous Checkpointing (DCP)** with dynamic topology resharding across mismatched GPU cluster sizes; verified across 24 unit test suites.

14. AutoCut-AI: Autonomous Multi-Modal AI Video Editor & Reel Generator ↑ Nav — **Aug 2026 – Present**
Multi-Modal Video Intelligence, Whisper Diarization, FFmpeg GPU NVENC & Next.js — GitHub: autocut-ai

- Engineered autonomous multi-modal video editing engine using **OpenAI Whisper diarization**, filler-word removal, and silence stripping; automated viral hook identification and generated 9:16 short-form reels in **<45 seconds (8x faster than manual editing)**.
- Built dynamic word-by-word animated subtitles using **FFmpeg GPU hardware acceleration (NVENC)** and computer vision face-tracking; achieved **4.2x viewer retention lift** across 100K+ generated social clips.

15. Nimbus-AI Suite: 5-Agent Live Conversational Voice Ecosystem (From Scratch) ↑ Nav — **Sept 2026 – Present**
LiveKit WebRTC, FastAPI, OpenAI GPT-4o / Whisper, ElevenLabs Turbo v2.5, RAG vs RAG-less & 60fps Barge-In — Live: nimbus.kishor.tech / GitHub: nimbus-voice-agents-suite

- Architected from-scratch **5-agent real-time voice platform** on **LiveKit WebRTC, FastAPI (/api/voice/process), OpenAI GPT-4o/GPT-4o-mini & Whisper ASR, and ElevenLabs Turbo v2.5 TTS**; achieved **sub-400ms end-to-end voice roundtrips** with live **60fps audio waveform visualization**, instant barge-in interruption, and latency waterfall telemetry.
- **Sales (sales.kishor.tech | GitHub) & Scheduling (scheduling.kishor.tech | GitHub):** Deployed autonomous SaaS advisor and quote checkout agents (annual savings calculator) alongside conflict-aware calendar coordination and automated reminders.
- **Meeting (meeting.kishor.tech | GitHub) & Research (research.kishor.tech | GitHub):** Built live meeting scribe extracting action items & minutes paired with deep-corpus literature synthesis, citation tracking, and swappable RAG vs. RAG-less retrieval.
- **Desktop (desktop.kishor.tech | GitHub):** Engineered hands-free OS voice automation for application control, system telemetry, and guarded scripts across all 5 Vercel microservices verified with automated `pytest` suites.

ACHIEVEMENTS & HACKATHONS

- **Competitive Programming & DSA:** Ranked **232nd in Codeforces Div-2** (**Blue Coder**), **5★ on CodeChef**, **17th in IUPC**; Qualified in **Meta Hacker Cup Round 3**; 2000+ DSA, participated system design hackathon.
- **Champion | Technobeast AI Agents Hackathon:** Pioneered Qdrant vector optimization with negative embeddings, trajectory matching with transformers, and multimodal API divergence detection.
- **7th Rank | AI Security Hackathon:** Ranked in top tier for real-time prompt injection firewall (**97.2% accuracy**, **<30ms** latency).
- **National Merit Scholarships:** Conferred merit scholarships for 85th Rank in Dhaka Board (HSC) and 90th Rank in Dhaka Board (SSC).

LEADERSHIP & MENTORING

- **In-Person AI Agents Building Masterclass:** Conducted immersive hands-on in-person workshops on architecting autonomous AI Agents, cyclic state machines (**LangGraph**), tool calling, and production deployment.
- **Technical Mentorship & Community Impact:** Mentored **100+ students & engineers** across **Machine Learning, AI Agents, DevOps, and MLOps**, leading code reviews, production architecture design sessions, and career roadmapping.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C/C++, Rust (Actix-web), Go (Gin/Fiber), SQL, Bash, TypeScript, JavaScript, Java (Spring Boot), FastAPI, Express.js, NestJS, Next.js 16, Streamlit
- **AI, ML & LLMOps:** PyTorch, vLLM, Vapi, Deepgram Nova-2, Cartesia Sonic-3.5, LangGraph, LangChain, HuggingFace, LangSmith, LangFuse, DeepEval, TRL, PEFT (RSLoRA, QLoRA), NeFTune, MLflow, Feast, Kubeflow, AWS SageMaker, KServe, DVC
- **Big Data & Lakehouse:** Snowflake, Databricks, Azure Data Factory (ADF), Azure Synapse Analytics (CETAS), ADLS Gen2, dbt Core, PySpark, Delta Lake, Apache Kafka, Apache Airflow, Debezium CDC, DuckDB, ChromaDB, Qdrant
- **Cloud Platforms & Infra:** AWS (EKS, S3, KMS, EMR, Athena, ECS, Fargate, Bedrock, SageMaker), GCP (GKE, BigQuery, GCE, VertexAI, Gemini AI), Azure (Databricks, Synapse, Fabric, vLLM, AKS), Terraform, Ansible, Helm, ArgoCD, Sonatype Nexus, Docker, Kubernetes (GKE, EKS, K3s)
- **DevSecOps & Observability:** NVIDIA NeMo Guardrails, PyRIT RedTeaming, BotShield WAF, SonarQube, Trivy, Jenkins, GitLab CI/CD, GitHub Actions, Prometheus, Grafana, EFK (Elasticsearch/Kibana), Redis
- **WebRTC, 3D Graphics & Full-Stack Systems:** WebVR, WebRTC, Three.js, React Three Fiber, ElectronJS, Next.js 16, Clerk Auth, n8n, Twilio, Google Calendar API v3, Neon Serverless PostgreSQL, Prisma, Drizzle ORM, Stripe Connect, MongoDB Atlas

CERTIFICATIONS & SPECIALIZATIONS

- **Deep Learning Specialization (DeepLearning.AI & Stanford – Andrew Ng):** Neural Networks, Hyperparameter Tuning, CNNs, Sequence Models, Attention Mechanisms & Transformers.
- **Machine Learning Specialization (Stanford Online & DeepLearning.AI – Andrew Ng):** Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders & Reinforcement Learning.
- **Enterprise Production MLOps & AI Systems Architecture:** End-to-End MLOps Pipelines (Krish Naik/CampusX), Enterprise AI Systems Architecture (Vivian Aranha), Agentic-AI, n8n (Mayank Agarwal), Computer Vision (Sourangshu Paul), AI Data Analyst (Ansh Lamba), Automation & MCP (Ed Donner)
- **Distributed Systems, Big Data & AI System Design:** High-Throughput Distributed Systems & AI Architecture (Arpit Bhayani, Gaurav Sen, ByteByteGo, Hussein Nasser), Big Data Engineering & Lakehouses (Mayank Agarwal, Zach Wilson), LLM Foundations (Karpathy, Aurimas), LeetCode algorithmic mastery.
- **Digital Portfolios & Systems Architecture Showcases:** ai-voice-receptionist.kishor.tech | nimbus.kishor.tech | kishor.tech | pptx.kishor.tech | 3d.kishor.tech
- **Core CS & Engineering Foundations (RUET):** Advanced Data Structures & Algorithms, Object-Oriented Programming (C++/Java), Operating Systems & System Programming, Database Systems (DBMS), Distributed Systems & Cloud Computing, Computer Networks & Internet Protocols, Theory of Computation & Automata, Compiler Design, Software Engineering & System Analysis, Computer Architecture, Artificial Intelligence, Microprocessors & Interfacing, Discrete Mathematics, and Network Security & Cryptography.